

PROJECT BOATY

Subsystem Specification Mission Control

Be the only place people interact with Boaty: turn words into safe, approved missions, and show, say and record what the boat is doing.

Document	BOATY-SSS-MCN
Issue	Issue D (for review)
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Status	For review by the project owner
Parents	SRS Issue F, ADD Issue F, ICD Issue F, FMEA Issue E
Content	65 allocated SRS requirements → 64 subsystem requirements (45 M, 18 S, 1 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	28 September 2026	FMEA actions A-03, A-04, A-10, A-13, A-15, A-16, A-18, A-19: MCN-D53 to D60 added.	Claude, owner decisions (CR-02, FMEA actions)
C	28 September 2026	CR-03: adult PIN on the web UI replaces the panel key switch (MCN-D02, D14, D60, D61).	Claude, owner decision
D	29 September 2026	Simulator slice 3 (Mission Control built and flown on SITL): MCN-D29, D39, D45, D57 and D60 clarified from the evidence; MCN-D62 to D64 added.	Claude

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Be the only place people interact with Boaty: turn words into safe, approved missions, and show, say and record what the boat is doing.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Raspberry Pi 5, case and panel (4 buttons, LEDs) • USB Wi-Fi adapter and antenna (pole kit if needed) • USB mic and speaker • Application components C1-C10 • Site files, logs, captain's logs	• The phone/tablet (display, internet) • The Claude API • Helm behaviour (HLM)

1.3 Breakdown

Item	Description
MCN-1 Hardware	Pi 5, cooler, case, panel, radio, audio
MCN-2 Session & UI	C1 session manager, C2 panel I/O, C3 web UI, C4 voice
MCN-3 Mission chain	C5 planner, C6 validator, C7 helm interface
MCN-4 Records	C8 photos & log, C9 site store, C10 logger

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue D	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue C	Interface definitions (all listed in section 3 apply in full)
BOATY-FMEA-001 Issue B	Failure modes and actions carried into this issue

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-01 Boat-bank radio link	Mission computer	Owner	\$4
IF-02 Command & telemetry	Helm	Owner	\$4
IF-03 Photo & health service	Mission computer	Conforms	\$4
IF-10 Internet uplink	Phone	Owner	\$4
IF-11 Claude API	Claude API	Owner	\$4
IF-12 Crew & operator panel	Operator & crew	Owner	\$7
IF-13 Mission format	internal	Owner	\$5
IF-14 Helm API	internal	Owner	\$5
IF-15 Site data	Simulation & test	Owner	\$5
IF-21 Simulation link	Simulation & test	Conforms	\$5

4. Parent requirements

Primary (65): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
MOD-006	M	While armed, the operator shall be able to command HOLD, RTL, MANUAL or STOP from Mission Control at any time,...	MCN-D11
MOD-007	S	Every mode change shall be shown and announced at Mission Control within 2 s, whatever caused it.	MCN-D12, MCN-D60, MCN-D62
PRE-005	M	For AUTO: a mission that has passed validation and operator approval shall be loaded, and read back from the helm to...	MCN-D15, MCN-D43
PRE-006	M	The pre-launch checklist (OPS-001) shall be complete and signed off at Mission Control for this session.	MCN-D13
FEN-002	M	Fences shall be drawn on a map at Mission Control, stored per site with a name, and version-controlled.	MCN-D19, MCN-D53
FEN-003	S	Mission Control shall show a 5 m guide inside the drawn inclusion fence, and warn if any home or waypoint falls within...	MCN-D19
FS-009	M	STOP: pressing STOP at Mission Control shall stop both motors within 1 s, whatever the mode.	MCN-D10
SAF-004	M	Output from the language model shall never be sent to the helm as a command. Only mission objects that have passed...	MCN-D38
SAF-007	S	Safety parameters (fence, failsafe thresholds, speed limits) shall be version-controlled. The values loaded on the...	MCN-D45
MIS-001	M	The system shall support these mission elements: waypoint, area survey (lawnmower pattern over a polygon, with lane...	MCN-D34
MIS-002	M	Every mission shall end with return home.	MCN-D34
MIS-003	M	Planned mission duration shall not exceed 20 min (configurable up to 30 min). Planned energy shall not exceed 50% of...	MCN-D35
MIS-004	M	Missions shall be represented in a documented JSON schema with a version number.	MCN-D36
MIS-005	M	Offline templates shall be available without internet, at least: 'Explore the bay', 'Duck patrol' (survey with photo...	MCN-D37
MIS-006	S	An adult shall be able to create and edit missions on the map at Mission Control.	MCN-D20
NLI-001	M	Mission Control shall accept instructions as typed text.	MCN-D26
NLI-002	S	Mission Control shall accept spoken instructions through a microphone, using push-to-talk on the TALK button.	MCN-D25
NLI-003	M	Instructions shall be turned into a mission through the Claude API. The request shall include the site context: fence,...	MCN-D28
NLI-004	M	The language model's reply shall be constrained to the mission schema (structured output or tool use). A malformed...	MCN-D29
NLI-005	M	The proposed mission shall be shown on the map with a one-sentence plain-language summary, which shall also be spoken...	MCN-D30
NLI-006	M	Requests the boat must not or cannot do (for example leaving the fence, chasing animals, going to another lake) shall...	MCN-D31
NLI-007	S	With mobile data available, a plan shall be displayed within 20 s of the instruction.	MCN-D32
NLI-008	M	The API key shall be stored only on Mission Control and shall never be committed to the repository.	MCN-D33
VAL-001	M	Every mission, from any source, shall pass the validator before upload. There shall be no other path for uploading...	MCN-D38
VAL-002	M	Every waypoint, and every leg sampled at 1 m intervals or finer, shall be inside the inclusion fence by at least 3 m...	MCN-D39

SRS	Pri	Requirement (abridged)	Met by
VAL-003	M	The mission shall meet the duration and energy limits in MIS-003.	MCN-D39
VAL-004	M	The mission shall start within 10 m of home and end with return home.	MCN-D39
VAL-005	M	Commanded speeds shall be within NAV-003. Each hold shall be 60 s or less. The photo count shall fit in the free...	MCN-D39
VAL-006	M	The validator shall be pure Python with no network access and no randomness. It shall have a unit-test suite that...	MCN-D40
VAL-007	M	Each rejection shall be reported in plain language, naming the rule broken and where on the map.	MCN-D41
VAL-008	M	After validation, an adult shall approve the mission at Mission Control. GO shall stay disabled until then.	MCN-D08, MCN-D42, MCN-D56
VAL-009	M	Any change to an approved mission shall void the approval.	MCN-D42
VAL-010	M	After upload, the mission shall be read back from the helm and compared with the approved mission by checksum before...	MCN-D43, MCN-D64
DET-001	S	After a mission, the system shall identify photos containing waterfowl, with at least 80% precision on a test set of...	MCN-D48
DET-002	S	The system shall produce a captain's log: up to 6 best photos, a map of the route, distance, duration, ducks seen, and...	MCN-D49
MC-001	M	Mission Control shall run on the owned Raspberry Pi 5 and operate for at least 2.5 h from a USB-C power bank.	MCN-D01
MC-002	M	It shall have four physical push buttons, each at least 30 mm across, in distinct colours with icons: GO (green), COME...	MCN-D02, MCN-D55
MC-003	M	STOP and COME HOME shall always be active while armed. GO shall be active only when a validated, approved and verified...	MCN-D08
MC-004	S	GO shall require a 1 s press-and-hold, to prevent accidental starts.	MCN-D09
MC-005	M	The display shall show, updated at least once a second: a map with the boat's position, heading and track; fence,...	MCN-D16
MC-006	S	The display shall be readable in daylight. A tablet or phone used as the display is acceptable.	MCN-D17
MC-007	S	Mission Control shall give spoken announcements of mode changes, failsafes and mission events in child-friendly words.	MCN-D27, MCN-D62
MC-008	M	Adult-only functions (arming, fence editing, parameter changes, mission approval) shall be protected by a PIN or key...	MCN-D14, MCN-D61
MC-009	M	The last known position and track shall stay displayed after link loss, in a 'find my boat' view.	MCN-D18
MC-010	M	Mission Control shall host the pre-launch checklist (OPS-001) and record who signed it off, and when.	MCN-D13, MCN-D54, MCN-D55
MC-011	M	Everything except natural-language planning and cloud photo analysis shall work without internet access.	MCN-D07
MC-012	S	An adult shall be able to drive the boat in MANUAL from Mission Control, using an on-screen joystick or a gamepad.	MCN-D22
MC-013	S	The Mission Control case shall resist splashes (IPX4 intent).	MCN-D01
MC-015	S	Mission Control shall have a photo-review mode the crew can use: large thumbnails, stepped through with the buttons,...	MCN-D23
MC-014	S	QGroundControl or MAVProxy shall be usable as a backup ground station.	MCN-D24, MCN-D57
COM-001	M	The boat-to-bank link shall use licence-exempt 2.4 GHz Wi-Fi within UK power limits.	MCN-D03
COM-002	M	Link range shall be at least 100 m line of sight over water, with the boat antenna at least 150 mm above the water.	MCN-D03

SRS	Pri	Requirement (abridged)	Met by
COM-003	M	Within range: telemetry at 1 Hz or faster, and command latency of 1 s or less (95th percentile).	MCN-D46
COM-004	S	Link quality shall be shown at Mission Control and logged.	MCN-D21
COM-005	M	Mission Control shall reach the internet through a phone hotspot. Losing internet shall not affect the boat link.	MCN-D04
COM-006	M	The link shall be secured with WPA2 or better and non-default credentials. The boat shall accept commands only from...	MCN-D03
SWE-002	M	A Python helm interface shall provide: arm, disarm, set mode, upload mission, read mission back, start, hold, return...	MCN-D44
SWE-003	M	Mission-level code shall run unchanged against any helm implementation, selected by configuration: first ArduPilot...	MCN-D44
SWE-008	S	All components shall timestamp in UTC, taken from GNSS where available.	MCN-D06
SWE-009	S	Mission Control software shall start automatically within 60 s of power-on.	MCN-D05
LOG-002	M	Mission Control shall log: all commands, sign-offs and approvals; each mission as approved (JSON); language model...	MCN-D50
LOG-003	M	The logs shall be enough to replay any trip on a map, including every failsafe event.	MCN-D51
LOG-004	M	Photos shall be stored locally and never published automatically. Photos go to cloud services only if an adult enables...	MCN-D47
LOG-005	C	Logs and photos may be backed up at home and kept for at least 12 months.	MCN-D52
CHD-005	M	The crew controls (GO, COME HOME, STOP, TALK) shall be usable without reading, by colour and icon alone.	MCN-D02

Supporting (21): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

MOD-001, MOD-003, MOD-004, MOD-005, PRE-001, PRE-002, PRE-004, PRE-008, FEN-001, FEN-004, FEN-005, FEN-007, FS-012, SAF-001, SAF-002, SAF-003, CAM-007, CAM-008, SWE-004, OPS-001, CON-002

5. Subsystem requirements

Requirements are numbered MCN-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Hardware and platform

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D01	Mission Control shall be a Pi 5 (≥ 4 GB) with an active cooler in a printed splash-resistant case with drip edges. ≥ 2.5 h on a 20,000 mAh USB-C PD bank.	M	T	BENCH	MC-001, MC-013
MCN-D02	The panel shall have TALK, GO, COME HOME and STOP buttons (≥ 30 mm, colour + icon, LEDs), per IF-12. There is no panel key switch.	M	I	BENCH	MC-002, CHD-005, IF-12
MCN-D03	The radio shall be a USB adapter with an RP-SMA antenna, acting as AP per IF-01 (country GB, per-site channel, WPA2+, non-default passphrase).	M	T	LAKE	COM-001, COM-002, COM-006
MCN-D04	Internet shall come only via the phone USB tether per IF-10. The boat network has no route out.	M	T	BENCH	COM-005, IF-10
MCN-D05	The application shall start automatically ≤ 60 s after power-on.	S	T	BENCH	SWE-009
MCN-D06	The clock shall be disciplined from helm GNSS time when there's no internet. All logs are UTC.	S	T	BENCH	SWE-008
MCN-D07	Everything except natural-language planning and cloud photo analysis shall work offline, including cached map tiles per site.	M	T	BENCH	MC-011

5.2 C1 Session manager and C2 panel

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D08	The session state machine (ICD Figure 4) shall gate every button. GO works only in ARMED with a verified mission. COME HOME and STOP work in every armed state.	M	T	SIM	MC-003, VAL-008
MCN-D09	GO shall need a 1.0 ± 0.1 s press-and-hold.	S	T	BENCH	MC-004
MCN-D10	STOP shall call helm.stop() ≤ 50 ms after the button edge. End-to-end ≤ 1 s (IF-02 budget).	M	T	POOL	FS-009
MCN-D11	An adult shall be able to command HOLD, RTL, MANUAL or STOP at any time while armed, overriding AUTO.	M	T	SIM	MOD-006
MCN-D12	Mode changes shall be shown and spoken ≤ 2 s after the helm reports them. Auto-disarm after 60 s at home.	S	T	SIM	MOD-007, MOD-008
MCN-D13	The checklist shall hold every OPS-001 item and record who signed it off, and when. Arming is blocked until it's done.	M	D	SIM	PRE-006, MC-010
MCN-D14	Arming, approval, fence editing, parameter changes, manual drive and the cloud toggle shall need the adult PIN (IF-12: 6 digits, 10 min unlock, lockout after 5 failures).	M	T	SIM	MC-008, DD-17
MCN-D15	Before arming, Mission Control shall also check ≥ 8 satellites, a verified mission and ≥ 1 GB free locally.	M	T	SIM	PRE-005, PRE-001, PRE-008

5.3 C3 Web UI

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D16	The map view shall show boat, heading, track, fence, exclusions, home, mission, battery, link, mode and time left, at ≥ 1 Hz.	M	D	SIM	MC-005
MCN-D17	A high-contrast daylight theme (≥ 16 px text) shall work on phone and tablet.	S	D	LAKE	MC-006

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D18	The 'find my boat' view shall keep the last position, time and track after link loss.	M	T	SIM	MC-009
MCN-D19	The fence editor shall show the 5 m guide, warn on violations, save to the site file and commit it to git.	M	D	SIM	FEN-002, FEN-003, IF-15
MCN-D20	An adult map mission editor shall route every edit through the validator.	S	D	SIM	MIS-006
MCN-D21	Link quality (RSSI both ends) shall be shown and logged at 1 Hz.	S	D	POOL	COM-004
MCN-D22	Manual drive from an on-screen joystick or gamepad: 10 Hz MANUAL_CONTROL with a dead-man (release → zero).	S	D	POOL	MC-012
MCN-D23	Crew photo review: large thumbnails, GO = next, COME HOME = back to start, ducks highlighted.	S	D	SIM	MC-015
MCN-D24	With the application paused, QGroundControl shall be able to take over UDP 14550 as a backup ground station.	S	D	SIM	MC-014

5.4 C4 Voice

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D25	TALK held → record (≤ 15 s) → on-device STT → transcript shown to the adult. Audio is never stored or sent.	S	T	BENCH	NLI-002, V-07
MCN-D26	A typed instruction box shall be available on the web UI.	M	D	SIM	NLI-001
MCN-D27	Spoken announcements shall use the IF-12 phrase set via on-device TTS.	S	D	SIM	MC-007

5.5 C5 Planner

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D28	Requests shall follow IF-11 and include site context (names, sizes, battery, limits). No coordinates.	M	T	SIM	NLI-003, IF-11
MCN-D29	Replies shall be validated against Intent v1. On failure, retry once with the problem passed back as data (≤ 2 allowed), then offer templates. Planner and validator failures count the same way.	M	T	SIM	NLI-004
MCN-D30	The plan shall be drawn on the map, and its summary_for_child shown and spoken.	M	D	SIM	NLI-005
MCN-D31	Declines (off-site names, chasing animals, leaving the fence) shall come from the system prompt rules and planner checks, worded for a child, with an alternative offered.	M	T	SIM	NLI-006
MCN-D32	Plan display ≤ 20 s at the 95th percentile on 4G.	S	T	LAKE	NLI-007
MCN-D33	The API key shall live in a 0600 file outside the repository. CI runs a secret scan.	M	I	SIM	NLI-008
MCN-D34	The geometry generator shall make survey lanes (light 20 m, medium 12 m, thorough 8 m spacing), photo stops $\geq$ keep_out_m from landmarks, and a final RTL.	M	T	SIM	MIS-001, MIS-002
MCN-D35	The duration and energy estimate shall be conservative ($\times 1.2$) and enforce 20 min (max 30) and 50% of usable energy.	M	T	SIM	MIS-003
MCN-D36	Intent, Mission and ValidationResult shall be versioned pydantic models per IF-13.	M	I	SIM	MIS-004, IF-13
MCN-D37	At least three offline templates, parameterised by site areas.	M	D	SIM	MIS-005

5.6 C6 Validator and C7 helm interface

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D38	helm.upload_mission shall accept only a ValidatedMission type that only the validator can create (type-level single path).	M	T	SIM	VAL-001, SAF-004
MCN-D39	The validator shall implement VAL-002 to VAL-005 in local ENU metres, sampling legs every ≤ 1 m, including the final straight leg home. Each leg's ends are checked against the fence first, and only legs with both ends inside are sampled (a waypoint at 0° N 0° E once cost 56 s of sampling).	M	T	SIM	VAL-002, VAL-003, VAL-004, VAL-005
MCN-D40	The validator shall do no I/O and be deterministic, with $\geq 95\%$ branch coverage and ≥ 50 adversarial cases.	M	T	SIM	VAL-006
MCN-D41	Violations shall be reported with rule, adult and child messages, and a map location.	M	D	SIM	VAL-007
MCN-D42	Approval shall bind to the mission checksum. Any change voids it.	M	T	SIM	VAL-008, VAL-009
MCN-D43	GO shall be enabled only after read-back checksum equality (IF-02).	M	T	SIM	VAL-010, PRE-005
MCN-D44	C7 shall implement the IF-14 Helm protocol (ArduPilotHelm, pymavlink), selected by configuration.	M	T	SIM	SWE-002, SWE-003, IF-14
MCN-D45	On connect, C7 shall compare helm parameters with the committed file. A mismatch blocks arming and shows the difference. Values the parameter stream drops are fetched again by index, so link loss never shows as a false difference.	S	T	SIM	SAF-007
MCN-D46	Telemetry ≥ 1 Hz and command latency ≤ 1 s (P95) shall be measured and logged every session.	M	T	POOL	COM-003

5.7 C8 Photos & log, C10 logger

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D47	Photos shall be synced per IF-03 after each mission and sha256-verified before ack. Kept locally, never published automatically. Cloud analysis off by default per session.	M	T	BENCH	LOG-004, CAM-007
MCN-D48	The waterfowl finder shall reach $\geq 80\%$ precision on the lake test set (cloud when enabled, else on-device heuristic).	S	T	LAKE	DET-001
MCN-D49	The captain's log shall have ≤ 6 photos, route map, distance, duration, ducks and a child-friendly story, saved as HTML/PDF.	S	D	LAKE	DET-002
MCN-D50	The session log shall be structured JSON lines: commands, approvals, missions, LLM I/O, validator results, events, RSSI.	M	I	SIM	LOG-002
MCN-D51	After each session the helm log shall be downloaded. A replay tool shows the trip and every failsafe on a map.	M	D	SIM	LOG-003
MCN-D52	Sessions may sync to a home computer when on the home network, kept ≥ 12 months.	C	D	SIM	LOG-005

5.8 FMEA-driven (Issue B)

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D53	The site linter shall check the IF-15 rules, plus: every home is inside the inclusion fence; the site is within 1 km of its configured reference; coordinates are in [lon, lat] order and inside a UK bounding box. It runs on save and at session start, and a failure blocks arming.	M	T	SIM	FEN-002, IF-15, A-04, FM-11
MCN-D54	Checklist rail test: key out $\rightarrow$ motor rail < 0.5 V shown; key in $\rightarrow$ rail $\approx$ battery. Failure shows 'key switch fault' and blocks arming.	M	T	BENCH	MOD-003, MC-010, A-10, FM-22
MCN-D55	Checklist button test: TALK, GO, COME HOME and STOP each pressed, with LED and sound confirming. Failure blocks arming.	M	T	BENCH	MC-002, MC-010, A-13, FM-32

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D56	The plan preview shall show duration, distance, photo count and furthest point from home, prominently and before the approve control.	S	D	SIM	VAL-008, A-15, FM-36
MCN-D57	C7 shall detect any other system-255 heartbeat on the link. Until the other source goes it refuses every command that starts or continues motion (arm, upload, GO, manual, drive) and raises an alarm. STOP, HOLD and come home stay available.	M	T	SIM	MC-014, A-16, FM-41
MCN-D58	Site files list launch points with suitable wind directions (IF-15). The checklist asks 'Is the wind blowing towards us?' and suggests a launch point. 'No' blocks arming.	S	D	SIM	OPS-001, IF-15, A-19, FM-42
MCN-D59	C1 shall command HOLD if a fence breach persists > 30 s or the boat is > 10 m outside (interim path of HLM-D24).	M	S	SIM	FEN-006, A-18, FM-14
MCN-D60	Boat-service holds (B5, B6, B7) shall be shown and spoken with the reason, which comes from the service's event text (the mode change can arrive first). A HOLD nothing explains is announced after 1 s. Weed-shedding is spoken once per episode. Resuming requires the adult PIN.	M	T	SIM	MOD-007, A-03, A-08
MCN-D61	The PIN shall be entered only on the adult's device, never shown or spoken, and changeable in Settings. The owner changes it at the start of each season.	M	I	SIM	MC-008, FM-48

5.9 From the simulator (Issue D)

ID	Requirement	Pri	Ver	Stage	Trace
MCN-D62	C1 shall follow mission progress by item sequence: the final return-home item runs inside AUTO, so 'coming home' and photo points are announced from the sequence and position, not the mode. RTL mode during a mission means a failsafe or an adult command.	M	T	SIM	MOD-007, MC-007, IF-13
MCN-D63	C1 shall stop the motors with the boat still armed (IF-14 halt()) for MCN-D59, so an adult can still bring it home.	M	T	SIM	FEN-006, IF-14
MCN-D64	VAL-010 shall compare the helm view of the mission (kind, position, hold, speed; ICD IF-13 Issue F). Photo counts go to the camera service and are not in the read-back.	M	T	SIM	VAL-010, IF-13

6. Component map

Mission Control's application components (ADD section 4.3) and the derived requirements each implements. Button behaviour by state is defined in ICD IF-12.

ID	Component	Responsibility	Derived requirements
C1/C2	Session manager, panel I/O	States, gating, checklist, adult key	MCN-D08 ... D15
C3	Web UI	Map, editors, photo review, manual	MCN-D16 ... D24
C4	Voice	TALK, STT, TTS	MCN-D25 ... D27
C5	Planner	Claude intent, geometry, templates	MCN-D28 ... D37
C6/C7	Validator, helm interface	Single mission path, read-back, parameters	MCN-D38 ... D46
C8/C10	Photos & log, logger	Sync, finder, captain's log, replay	MCN-D47 ... D52
C9	Site store	GeoJSON sites (IF-15)	MCN-D19

7. Design constraints and budgets

- ADD AR-2: intelligence on the bank
- ADD DD-04: LLM returns intent; Python makes geometry
- ADD DD-06/07/08: Pi 5 AP, on-device speech, web UI

Budget	Allocation
Mass	Not constrained (bank)
Cost allocation	£26 (buttons £6, mic/speaker £8, adapter £12) + £8 pole kit if needed
Power	≈ 6.5 W average from a USB-C PD bank

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	24	15	4	-	1	44
BENCH	10	-	1	-	-	11
POOL	2	2	-	-	-	4
LAKE	3	2	-	-	-	5

Ref	Open item
TBC-10	Closed: TALK button (ICD Issue B)
V-07	On-device speech latency and child accuracy
V-08	Link range with and without the pole